Early Stress and Later Response to Separation in Rhesus Monkeys

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The hypothesis that separation in early life predisposes one to later psychopathology was tested using two groups of rhesus monkeys of similar age. The experimental group had undergone the stress of early life separation and confinement in a vertical chamber. The experimental animals responded to separation and reunion with increased self-mouthing, self-clasping, huddling, and rocking. The control group responded to separation with increased locomotion and responded to reunion with increased contact clinging and proximity behaviors. The implications of these findings are discussed.

SEPARATION EXPERIENCES in early life have long been regarded in psychiatric theory and practice as traumatic events that can predispose individuals to psychopathology later in life (1, 2). In general the validity of this relationship has not been questioned (3). However, while psychiatric evidence obtained through a retrospective anamnesis may well have established the importance of separation in early life to the psychological functioning of the adult, it has also been criticized for making it appear that certain sequelae proceed inevitably from a given event (4). We know, for example, that not every individual who experiences a difficult separation in infancy eventually develops serious psychopathology. It therefore behooves investigators to regard the relationship between adult psychopathology and trauma in early life as a hypothesis and to attempt to establish experimentally whether such a relationship exists. Some have suggested that separation in early life may "leave scars" that sensitize individuals to adverse reactions to later separation (5). The experiment reported in this paper is a controlled laboratory investigation of this hypothesis in nonhuman primates.

Various efforts have been made in recent years to supplement the retrospective method in human beings with other types of studies. For example, Sethi obtained information on both childhood separations and separation experiences occurring within six months prior to the onset of depression in 116 patients (6). He found that 63 percent of the patients who scored high on a clinical inventory of depression had had a history of antecedent separation, whereas only 14 percent of those who scored low on the inventory had had such an experience. On the same inventory, 67 percent of the patients who scored high had had a separation experience before the age of 16, whereas only 37 percent of those who scored low had had such an experience. Finally, 44 percent of the patients who scored high had had *both* early and recent separation experiences, compared with 11 percent of those who scored low.

In a group of 120 hospitalized patients, Levi found significantly more incidents of separation before the age of 16 in patients who actually attempted suicide than in those who did not attempt suicide (7). The data in this study also suggested that there was a significantly higher incidence of separation experience before the age of seven in persons who actually attempted suicide than in those who only had an impulse to commit suicide.

Hill and Price compared 1,483 depressed patients with 1,028 nondepressed patients (8). They found a significantly higher incidence of loss of father by death before the age of 15 in the depressed group. A number of additional studies have found some positive relationship between the early loss of a parent and psychiatric disturbance in later life (9–11).

Birtchnell (12), in comparing 500 hospital admissions with a matched sample of 500 persons from the surrounding community, found a higher incidence of parent death before the age of five in the patient population. He also found a higher than expected incidence of recent parent death in the hospital population (13). Birtchnell failed to find a preponderance of early or recent parent death in patients with a diagnosis of depression, although these events were significantly higher in the subpopulation with *severe* depression (14). Moreover, his statistics indicated that early and recent parent death contributed to the incidence of suicide attempts in patients, independently of a diagnosis of depression (15).

Not all studies have indicated a positive relationship between early life events and psychiatric disturbance in later life (16–18). For example, Hudgens compared 40 depressed patients with 40 closely matched controls and failed to show any differences in the incidence of stressful events in early life (19).

Read in part at the 125th annual meeting of the American Psychiatric Association, Dallas, Tex., May 1–5, 1972.

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This work was supported by Public Health Service grants MH-11894 and MH-18070 and Research Scientist Development Award MH-4753 (to Dr. McKinney) from the National Institute of Mental Health.

The authors wish to express their appreciation to Elaine Moran and Robert Eising for their assistance with this project.

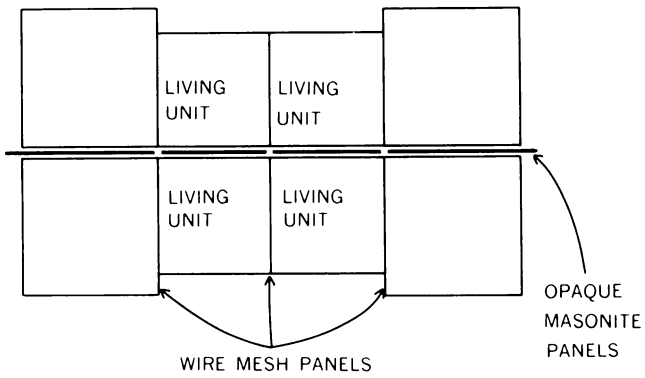
Other studies have indicated that the relationship between early life events and psychopathology is very complex. Paykel and associates have suggested that events of an "exit" type contribute more significantly to depression than do "entrance" events (20). A recent study by Cadoret and associates (21) comparing two groups with unipolar depression showed that, although there was no significant difference between the two groups in the actual incidence of such separation experiences as death, divorce, or separation of the parents, those patients whose onset of symptoms occurred before age 40 and who had certain other distinguishing characteristics showed a higher incidence of the *reporting* of a loss.

Many clinicians have established through intensive direct observation that dramatic behavior changes can occur in infants and young children who have undergone a major separation experience (22–24). Freud and Dann (25) studied a group of six children over a three-year period after their release from a concentration camp. Initially the children showed a high degree of attachment to each other and hostility toward the adults around them. It was only after several months that more typical child-child and child-adult behaviors began to appear. Spitz described a syndrome of "anaclitic depression" occurring in 19 of 123 children between six and 11 months of age in a foundling home (26). He postulated that this syndrome was caused by separation from the mother and could become irreversible if it was not remedied by replacement of a suitable mother or surrogate.

Bowlby characterized the human infant's response to maternal separation as occurring in three stages: "protest," "despair," and "detachment" (3). Such observations imply that loss in infancy might affect the adult individual's response to subsequent separation, but such follow-up studies are rare and give scattered results. Bowlby and associates (27) collected data on 60 children aged six to 14 hospitalized in a tuberculosis sanitarium for periods of several months and found very little indication of severe emotional impairment in adult life. Other investigators, however, on the basis of observations of institutionalized children (28, 29), have found evidence for lasting intellectual and emotional impairment that persisted into adolescence. Such studies are confounded by a complexity of variables, such as the age at separation, parent-child relationships before and after the separation, and the relative conditions of the environments (30).

In recent years some effort has been made to circumvent the difficulties involved in human clinical studies by creating animal models for human psychopathology (31–33). Nonhuman primates provide a useful model for some aspects of human behavior because of their closeness to man, their tendency to form close social bonds, the existence of a fund of knowledge about their social behavior, and the availability of suitable laboratory methods for well-controlled longitudinal studies. Early work using rhesus monkeys established the importance of the infant's tie to its mother (34). Experimental separation from the mother (35) produced a syndrome in rhesus monkeys that closely resembled the behavior of human infants described by Spitz and Bowlby. Replication of the separation experiment with other species has demonstrated a similar response to separation, with some species variation (36, 37). In the rhesus monkey, the species used in this study, the "protest" stage is characterized by frequent cooing vocalizations and increased locomotion. This stage is followed by a period of "despair," consisting of decreased locomotion and increased self-clasping, self-mouthing, and huddling. The final stage of "detachment" described in humans by Bowlby has not yet been reproduced in monkeys.

FIGURE 1
Cage Assembly of Experimental Living Unit



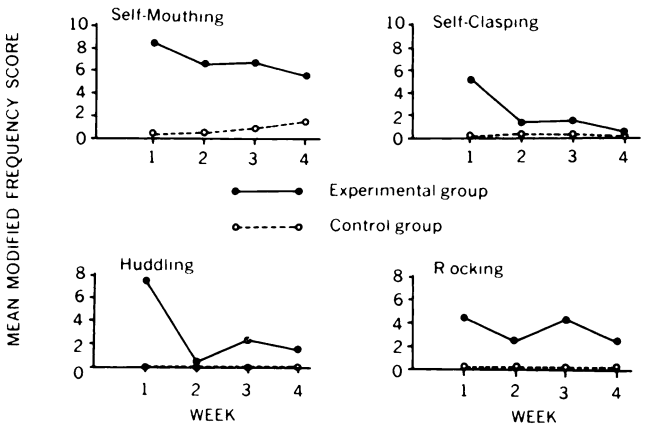
Separation from the mother before one year of age and confinement in a Harlow vertical chamber is a method that has proved effective and reliable in producing abnormal behavior in infant monkeys (38, 39). For example, monkeys 45 days of age placed in this chamber exhibit markedly abnormal behavior on removal. They are socially withdrawn, have low levels of locomotion, and exhibit high levels of self-mouthing, self-clasping, and huddling behavior.

This study attempts to assess whether such stress in early life predisposes rhesus monkeys to pathological behavior later in life when they are subjected to separation stress. Subjects who had early life separation and vertical chamber confinement were compared with subjects who had had normal laboratory rearing. The hypothesis was that these "traumatic" events early in life would predispose the subjects to more severe reactions to the stress of separation later in life.

METHOD

Two groups, each consisting of four rhesus monkeys almost two years of age, were used as subjects. The two groups had had different rearing experiences. The monkeys in the experimental group had been taken from their mothers at birth and housed in individual cages with cloth surrogates. They had been permitted two hours of peer experience per day throughout the first year of life. Between five months and ten months of age each subject

FIGURE 2
Self-Mouthing, Self-Clasping, Huddling, and Rocking During Separation



was placed in a vertical chamber (40) for 30 days of profound environmental isolation. Following the period of internment each subject was returned to its living cage. Beginning at one year of age, the experimental subjects were housed as a group in a wire mesh cage, where they remained until the start of the present experiment. In summary, they had undergone experimental early life separation and the stress of confinement in a vertical chamber early in life.

At the start of the experiment the subjects in the control group were approximately the same age as those in the experimental group. They had been reared with their mothers for the first month of life. From one month until one year of age they had lived in a large cage complex with their mothers, with two hours of daily social interaction with each other. From one year of age until the start of the experiment they were housed together without their mothers as a group of four. Therefore they were matched with the experimental group in terms of age and peer experience after one year of age but differed regarding early life separation and vertical chamber experience.

During the present experiment each group of four monkeys was housed together in a large experimental living unit, a series of cages joined end-to-end, in which the animals could both live and be tested without being moved to a strange environment. The cage assembly is shown in figure 1. The spaces between the cages permitted insertion of wire mesh partitions that divided each large unit into four smaller cages. When the wire mesh panels were removed, the open ends allowed free access through the entire length of the unit. Opaque masonite panels were placed between the two large cage assemblies to prevent visual and tactile communication between the experimental and control groups.

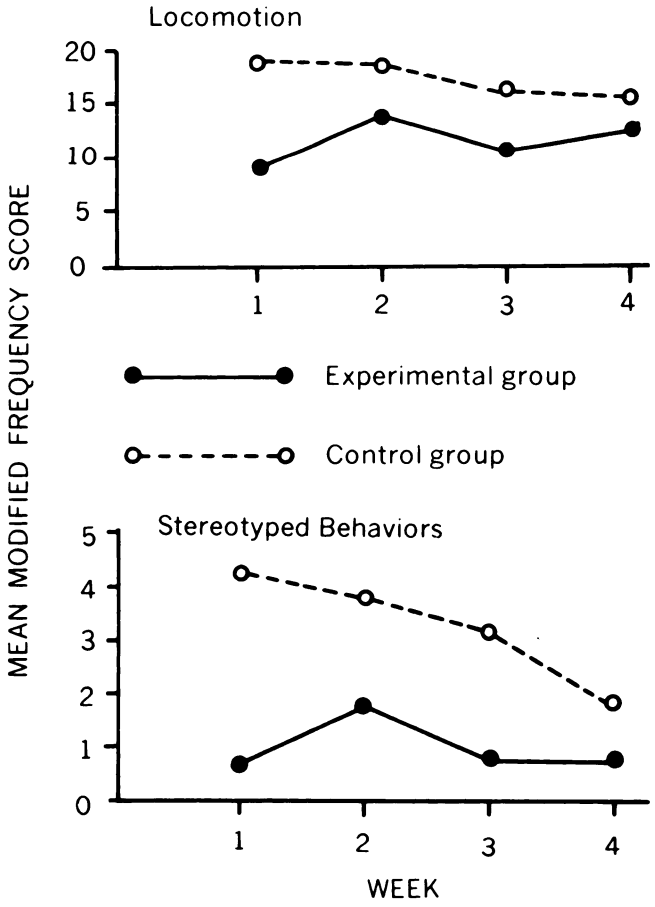
The experiment was divided into two time periods: a four-week period of baseline observation, called the preseparation period, followed by a four-week experimental period. During the preseparation period observations were made on four days each week while the monkeys had free access to each other and to their half of the entire cage assembly.

The experimental period was further divided into separation and reunion conditions. The subjects were housed individually 23 hours per day under the separation condition and were together one hour per day under the reunion condition on the days of observation. They were housed individually all day long on the remaining days.

During the separation condition the wire panels were slid into place so that each monkey was separated from his peers. Observations were made four times per week on each of the monkeys housed in this manner. Under the reunion condition the wire mesh panels were removed so that the monkeys had free access to each other and the full cage assembly. Observations were also made on the monkeys in this condition four times per week.

Behavioral observations were made on each animal for a five-minute period each morning on four days per week during the preseparation period. Similar five-minute observations were made of each subject as it was individually housed in the morning during the separation period and in the afternoon during the reunion condition. The five-minute observation periods were divided into 20 15-second intervals. Any of the operationally defined behavior categories that occurred during these intervals were scored as modified frequencies, that is, as the number of

FIGURE 3
Locomotion and Stereotyped Behaviors During Separation

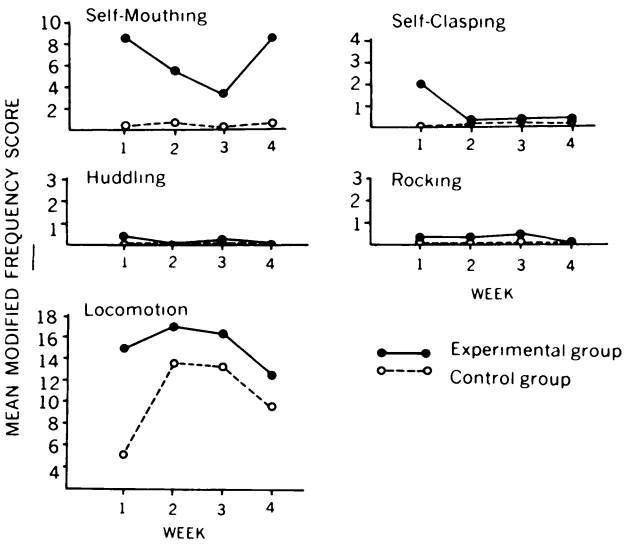


15-second intervals in which a behavior occurred. The use of this scoring system has been described previously in some detail (41). Operational definitions of the behavioral categories used are given in appendix 1.

For each animal a weekly mean modified frequency score was calculated thus: the total weekly number of responses of a given subject in a given category divided by four, the number of observation periods in the week. Behavior categories that consistently had scores of zero or nearly zero were omitted from the remaining analysis. For each of the remaining categories a two-way repeated measures analysis of variance was performed for each of the three experimental conditions (preseparation, separation, and reunio..). The prior rearing condition (experimental versus control) was the independent variable. Time blocks of two weeks for the preseparation condition and blocks of one week for the separation and reunion conditions were the repeated measures. The .05 level of confidence was designated as the criterion of statistical significance for each analysis. In those cases in which significance was reached or almost reached and more than two means were involved in the calculation of the mean square, a Duncan New Multiple Range test was used to compare the means with each other (42).

FIGURE 4

Self-Mouthing, Self-Clasping, Huddling, Rocking, and Locomotion During Reunion



RESULTS

Preseparation Condition

Four behavioral categories showed significant group effects during the preseparation condition. These were self-mouthing, locomotion, stereotyped behavior, and yielding. None of the other categories showed significant differences between the groups during this condition. The experimental group showed consistently higher levels of self-mouthing ($p < .01$). The control group showed higher levels of locomotion, stereotyped behavior, and yielding ($p < .05$).

Separation Condition

Under this condition, since the animals were alone, six social behaviors logically could not occur: yielding, contact cling, proximity, social exploration, social sex, and play. These were, therefore, excluded from the analysis of this condition. The experimental group showed significantly higher levels of self-mouthing and huddling ($p < .05$) during separation. The experimental group also showed higher levels of two other behaviors that did not reach significance—self-clasping and rocking. For self-mouthing the difference between the groups for the first week was significant by the Duncan test ($p < .05$). For rocking, the between-group differences for all four weekly means were significant by the Duncan test ($p < .01$). These differences are shown in graphic form in figure 2.

Self-clasping, huddling, and rocking were almost never shown by the controls. On the other hand, the control group showed higher levels of locomotion ($p < .05$) and stereotyped behavior ($p < .01$) during the separation condition. These differences are illustrated in figure 3.

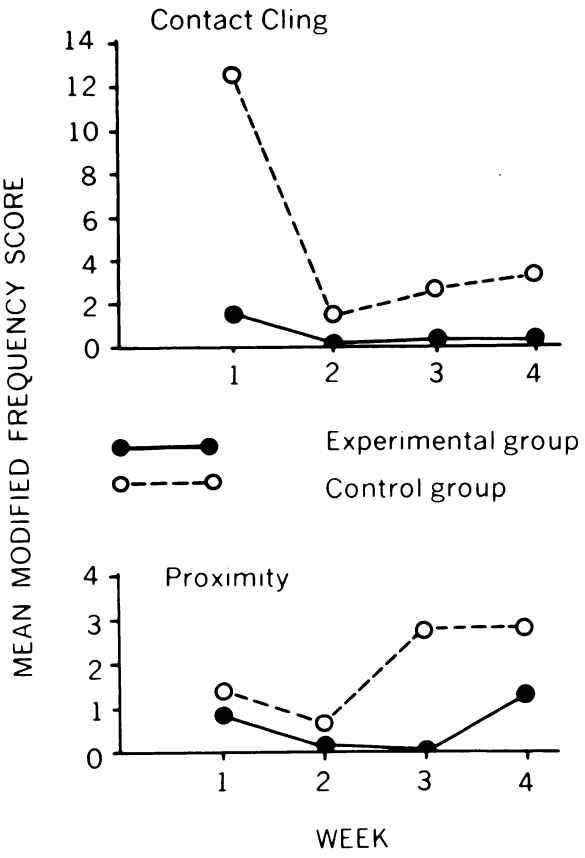
Reunion Condition

There were significant group differences for the following four behaviors, with the experimental group showing higher levels during reunion: self-mouthing and locomotion ($p < .01$) and rocking and self-clasping ($p < .05$). The experimental group also displayed some huddling, which was absent among the controls. The behavior categories for reunion in which the experimental group showed higher levels are illustrated in figure 4. The group means for proximity and contact cling behaviors are illustrated in figure 5. During reunion the control group showed significantly more contact clinging and proximity behaviors than the experimental group.

DISCUSSION

The results of this experiment indicate that several behavioral distinctions can be made between a group of four two-year-old rhesus monkeys with a history of early life separation and vertical chamber confinement and a group of monkeys raised with their mothers and with peers. Throughout all conditions of the experiment—preseparation, separation, and reunion—a higher rate of self-mouthing behavior was observed among the experimental group than among the controls. This tendency toward self-orality has been noted before in rhesus monkeys raised without mothers from birth (43). Self-mouthing is also commonly observed in infant rhesus monkeys immediately following vertical chamber confinement (44). The observation of high rates of this behavior under varying experimental conditions in older monkeys who have had separation and chambering experiences very early in life, and the virtual absence of self-

FIGURE 5
Contact Cling and Proximity During Reunion



mouthing in monkeys of similar age who have not had such experiences, indicates that self-mouthing may be a stable behavioral trait related to early history.

The experimental imposition of conditions of separation and reunion revealed further behavioral differences between the two groups that appeared to be labile and dependent on the condition. Separation conditions caused the experimental animals to react with dramatic increases in self-clasping, rocking, and huddling, behaviors that were almost never seen in the control group. Such "despair-like" responses are similar to those reported in infant macaques undergoing maternal separation and vertical chamber confinement for the first time (44). The pattern resembles the stage of "despair" described by Bowlby. In some respects, therefore, our imposition of the stress of separation from peers in these older monkeys appears to have "unmasked" or at least intensified a response to stress that probably developed in very early life.

The group of monkeys raised with mothers and peers, on the other hand, showed none of the "despair-like" behaviors of the experimental group. These animals responded to separation with an increase in locomotion and stereotyped behavior. This response is similar to that observed in monkeys reared with mothers and peers who are separated for the first time at three years of age (45). The dramatic response in the control animals was observed under reunion conditions. On being allowed access to each other again after separation, the control group showed higher levels of contact clinging and proximity behavior than the experimental animals. This response has been reported previously in animals undergoing reunion after separation, and it might be regarded as a typical pattern of response to reunion for juvenile monkeys raised with mothers and peers (46). The experimental group, on the other hand, did not respond with such high levels of these "closeness" behaviors, but rather with a higher level of locomotion than the control group. It appeared that separation and chambering inhibited their capacity to respond to reunion in the usual way.

The above findings are best viewed as tentative, we hope provocative, and suggestive for further research. Additional work is necessary to extend the generality of our hypothesis. Certainly, however, the marked tendency of rhesus monkeys with a history of maternal separation and vertical chamber isolation early in life to respond to separation with behaviors remarkably like those displayed in early life is supportive of our hypothesis. It remains to be seen whether such behaviors will appear in animals with similar histories at more advanced stages in life. In future studies, more emphasis will be placed on the important effect peer groups have on the final behavioral response, and appropriate measures will be taken to control for this. Work is currently under way in our laboratory to determine to what extent other forms of separation stress will tend to recapitulate the "despair-like" behavioral responses in monkeys with a history of early life stress.

Certainly it is difficult to analogize directly to human behavior but our findings may point to fruitful areas for exploration. The remarkable similarity of the syndrome produced in older monkeys by chambering to that displayed when they were young should cause us to look for behavioral similarities in follow-up data on adults who suffered clearly traumatic manifestations as a result of separation or environmental deprivation in childhood. Few studies of this sort have been reported (30). Our data indirectly support previous studies in human subjects that show a relationship between human psychopathology and the combination of separation in early life and recent separation (6, 12–15).

It appears that early and later life experience can combine to have a powerful influence on primate behavior, in man as well as monkeys. What remains to be decided is not whether such a relationship exists but how it comes to be.

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APPENDIX 1

Definitions of Behavioral Categories

Self-mouthing: Oral contact with any part of the body except the genital area, excluding biting.

Self-grooming: Discrete self-directed picking and/or spreading of the fur.

Self-biting: Specific, vigorous, self-directed biting.

Self-clasping: Firm manual or pedal clutching of self, excluding the genital area.

Passivity: Maintaining a static position for a minimum of three seconds with no other simultaneous behavior.

Huddling: Self-enclosed fetal-like position incorporating any or all patterns of self-clasping, self-embracing, or lowered head.

Rocking: Repetitive, nonlocomotive backward and forward movement.

Vocalization: Coo—drawn-out, dove-like cry in a variety of pitches; screech—high intensity vocalization resembling a scream.

Yielding: Oriented movement or withdrawal by an animal submitting or ceding area to another animal.

Stereotyped behavior: Identical movements maintained in a rhythmic and repetitive fashion for at least three cycles; scored independently and noninclusive of locomotion.

Locomotion: Ambulation of one or more full steps.

Environmental exploration: Any tactual or oral exploration of the physical surroundings.

Proximity: Standing within three inches of another monkey for a period of three or more seconds.

Contact cling: Maintaining gross body contact for an indefinite period of time.